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CS3002 Information Security

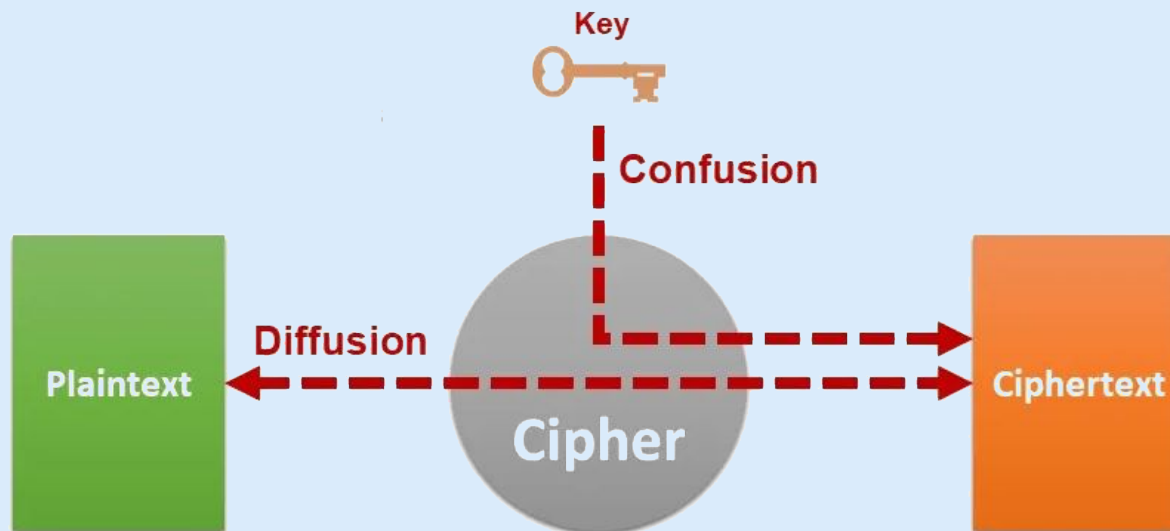


Modern Symmetric Ciphers

Strength of a cipher



- A strong cipher needs to completely obscure statistical properties of original message
- In 1945, Shannon suggested that a cipher needs to have these two properties:



Symmetric Crypto: How?



- There are two primary techniques

(1) Substitution – it creates “Confusion”



Replace each character (or bit string) with another character (or bit string)

(2) Permutation – it creates “Diffusion”



Change the order of characters or bit strings.

Confusion



- Confusion means that each character (e.g. bit) of the ciphertext should depend on several parts of the key, obscuring the connection between the two.
- This property makes it difficult to find key from ciphertext. If a single bit in key is changed, values of most or all bits in ciphertext will be affected.
- Confusion basically means that the equations that represent the ciphertext are too complicated and cannot be worked with.

Diffusion

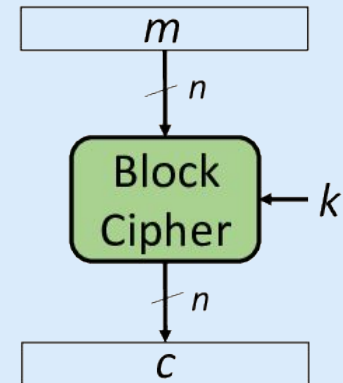


- Diffusion means the influence of one plaintext symbol is spread over many ciphertext symbols, so that statistical properties of plaintext remain hidden.
 - This will frustrate the adversary who uses ciphertext statistics to guess the plaintext.
- Diffusion implies that each symbol (character or bit) in the ciphertext is dependent on some or all symbols in the plaintext.
 - Conversely, if a single symbol in the plaintext is changed, several or all symbols in the ciphertext will also be changed.

Block & Stream ciphers



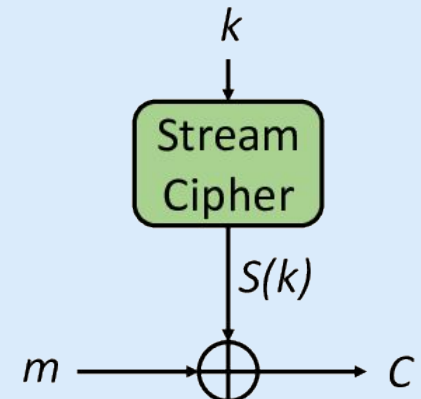
- **Block ciphers** operate only on messages of predetermined sizes – e.g. 64 bits or 128 bits.
 - Cipher encrypts one block of plaintext, creating a ciphertext block of same size.
 - To encrypt data smaller than block size, some kind of padding is essential.
 - Similarly, data larger than block size will need splitting.
 - Each character (e.g. bit) contributes to encryption of other characters in the block



Block & Stream ciphers



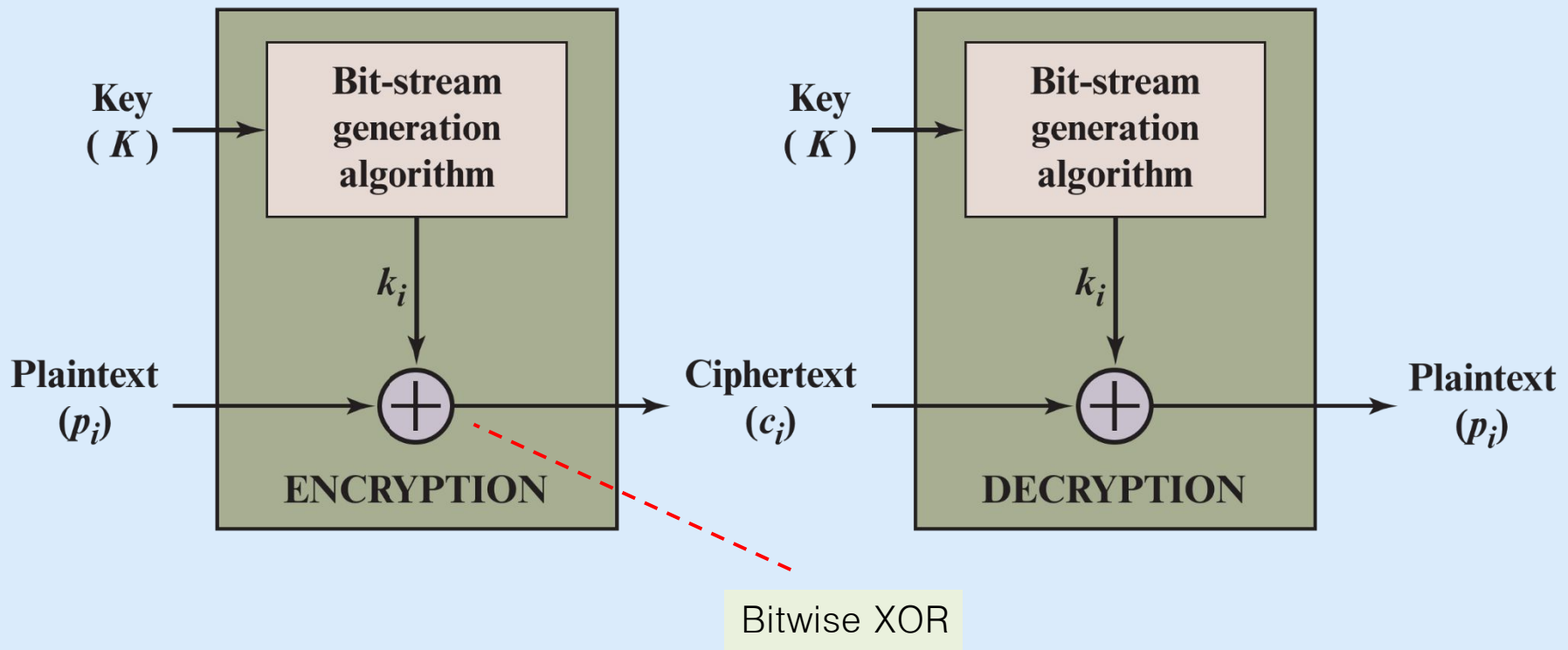
- In **stream ciphers** , encryption is done on much smaller unit (bit, byte etc.)
 - Units in plaintext are fed into encryption algorithm as a stream (one unit at a time), ciphertext is similarly outputted as a stream.
 - They are more suitable when data is a continuous stream



Block & Stream ciphers



Stream ciphers high level structure



XOR operation: extremely useful in crypto



XOR

0	0	0
0	1	1
1	0	1
1	1	0

AND

0	0	0
0	1	0
1	0	0
1	1	1

OR

0	0	0
0	1	1
1	0	1
1	1	1

Information
Leak!

- Primary reason: It does not reveal any information about plaintext (both 0 and 1 are equally likely).
- Secondary advantage: Both encryption and decryption can be done the same way, by adding the key. XOR is a self-inverse function!

$$\begin{array}{ccccc} 010010 & \oplus & 101100 & = & 111110 \\ \text{plaintext} & & \text{key} & & \text{ciphertext} \end{array}$$

$$\begin{array}{ccccc} 111110 & \oplus & 101100 & = & 010010 \\ \text{ciphertext} & & \text{key} & & \text{plaintext} \end{array}$$

Modern Encryption Algorithms



Examples

- Block Ciphers
 - DES
 - 3DES
 - AES
 - Twofish
- Stream Ciphers
 - RC4
 - ChaCha20

Block Ciphers

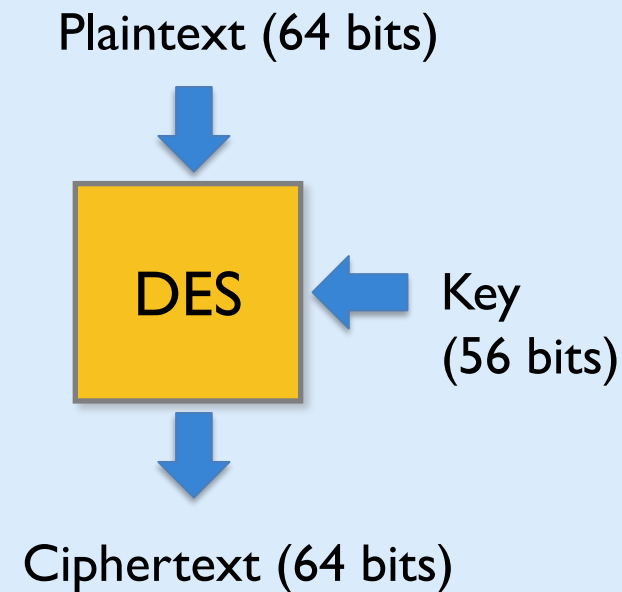


- “A modern block cipher can be designed to act as a substitution cipher or a transposition cipher. This is the same idea as is used in traditional ciphers, except that the symbols to be substituted or transposed are bits instead of characters.”
- A **product cipher** is the one which includes multiple rounds of both these operations (substitution, permutation).

Data Encryption Standard



- DES was once the most widely used encryption scheme
 - developed in early 1970s at IBM, slightly tweaked by NSA
 - the first modern, public, freely available encryption algorithm
 - Standardized by National Bureau of Standards in 1977
- Uses 64 bit plaintext block and 56 bit key to produce a 64 bit cipher text block



DES Strength Analysis



- DES is probably the most-scrutinized encryption algorithm, and to date no substantial mathematical weakness has been discovered in it.
- BUT... it is not resistant to brute force attacks due to short key length.

DES Strength Analysis



Short key length

- With 56 bits, there are 2^{56} possible keys (approx. 7.2×10^{16}), which can be brute forced in relatively short time.

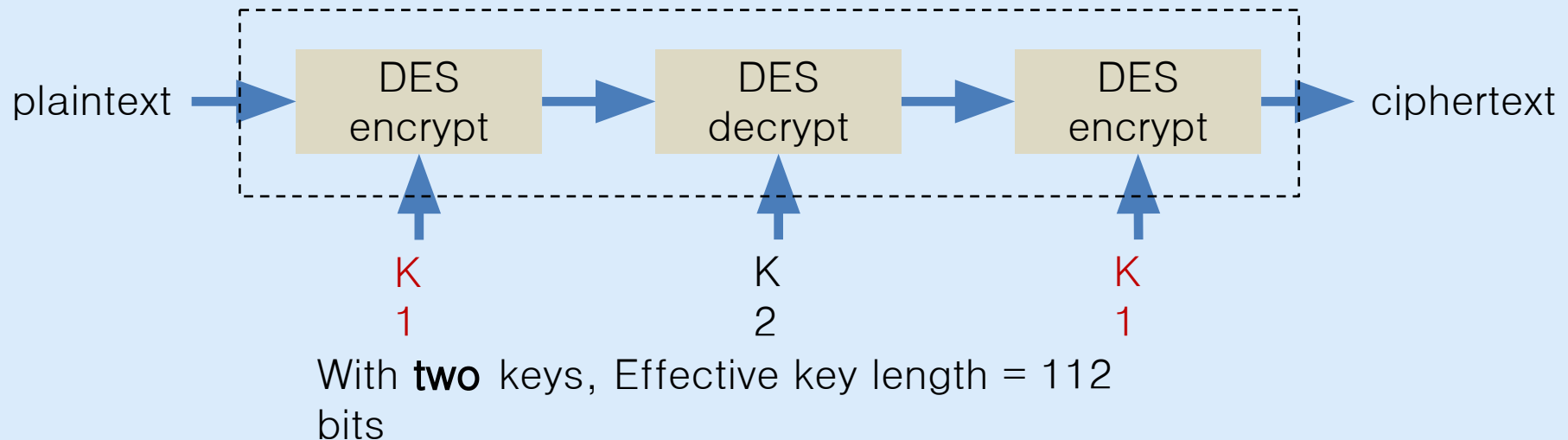
Chronology of DES Cracking	
Broken for the first time	1997
Broken in 56 hours	1998
Broken in 22 hours and 15 minutes	1999
Capable of broken in 5 minutes	2021

- More powerful and parallelized hardware yields quicker results
- Within 26 hours when using a specialized hardware:
<https://crack.sh>

Triple DES (3DES)



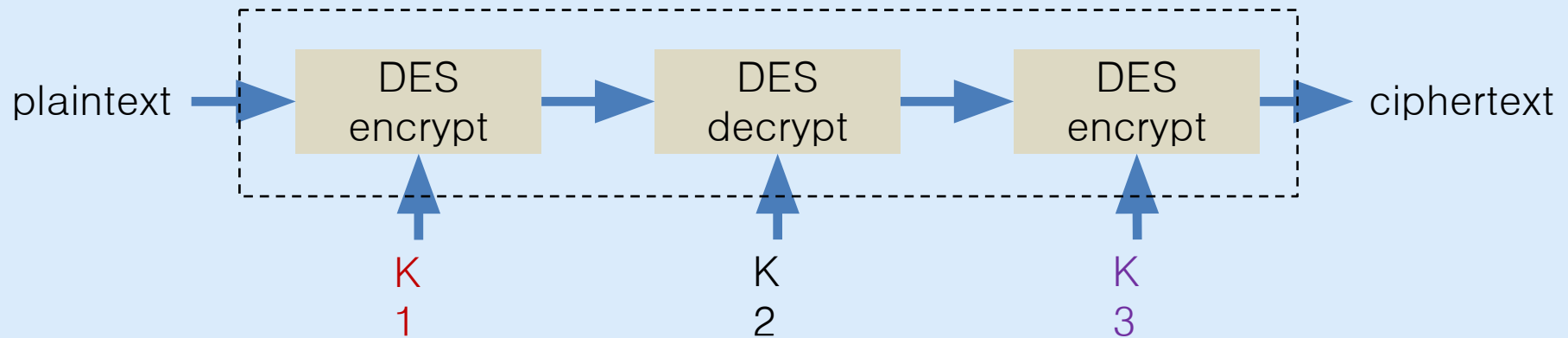
- Repeats basic DES algorithm three times
- Using either two or three unique keys
 - key size of 112 or 168 bits.
- much more secure but also much slower



Triple DES (3DES)



- Repeats basic DES algorithm three times
- Using either two or three unique keys
 - key size of 112 or 168 bits.
- much more secure but also much slower



With **three** keys, Effective key length = 168 bits

Block Cipher Modes of Operation

Block ciphers operate only on small fixed size chunks like 64 bits (DES), 128 bits (AES) etc.

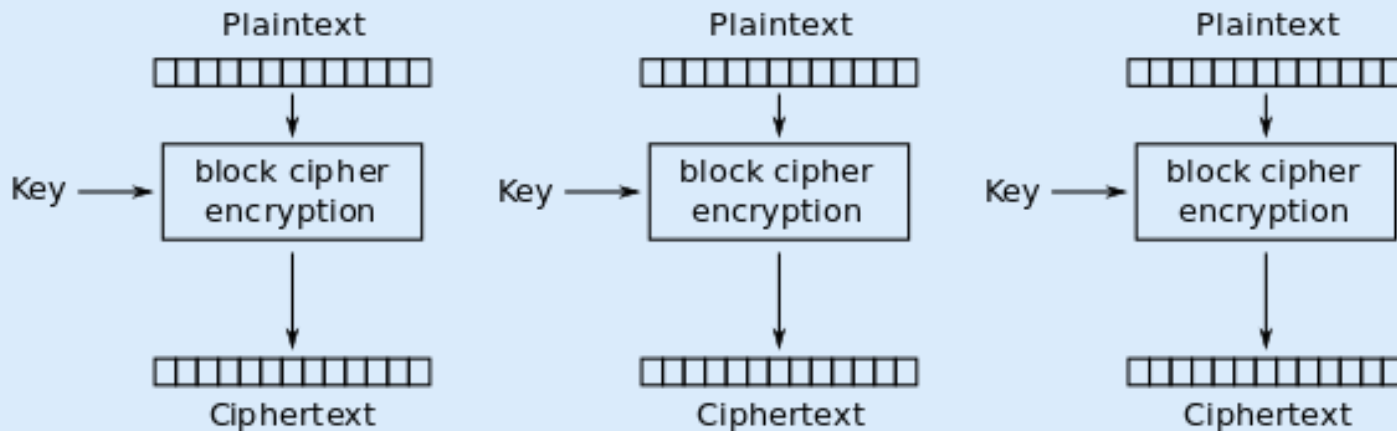
But in real world applications, we would need to handle arbitrary sized plaintexts.



Electronic Codebook mode

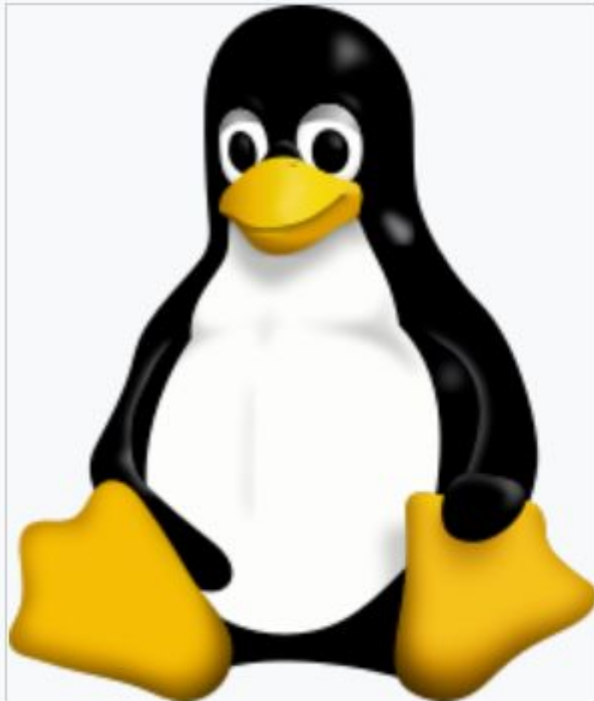


- To encrypt large data, one (lazy) option is to simply divide the whole plaintext in blocks and encrypt them separately with same key. This is ECB mode.



Electronic Codebook (ECB) mode encryption

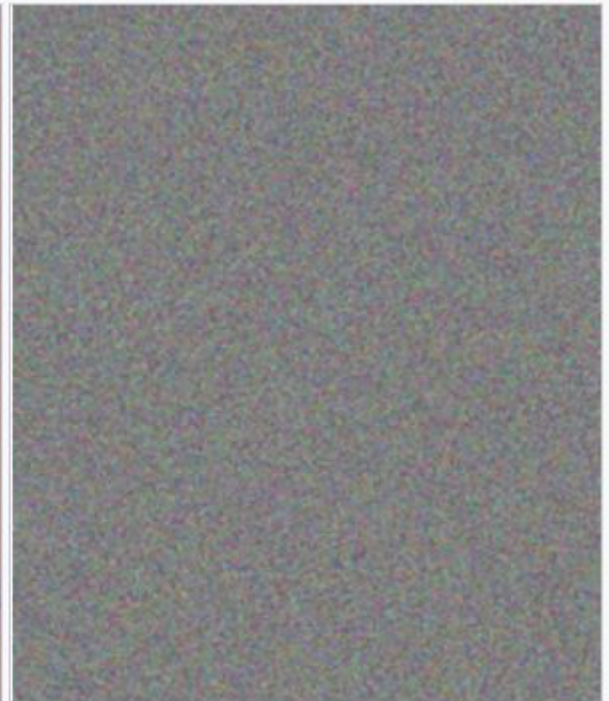
ECB mode: problem



Original image



Using ECB allows patterns to be easily discerned

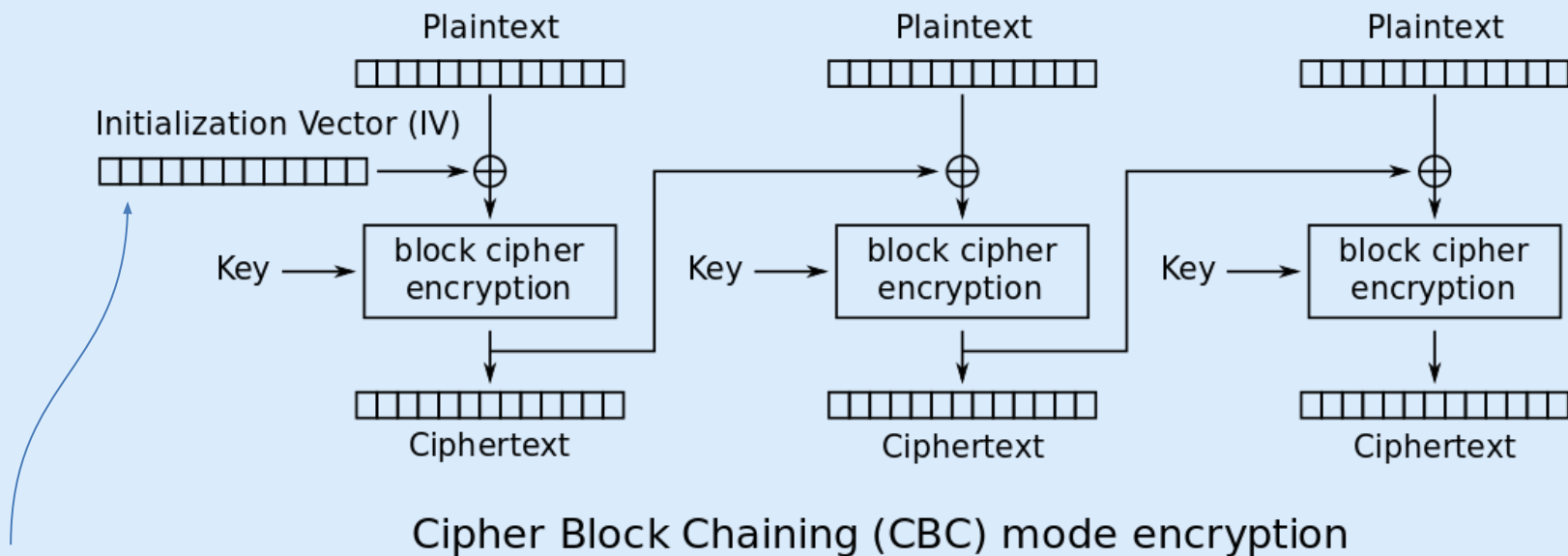


Modes other than ECB result in pseudo-randomness

Cipher Block Chaining mode



- More secure modes are available.
- In CBC mode, each ciphertext block depends on all previous plaintext blocks that have been processed up to that point.

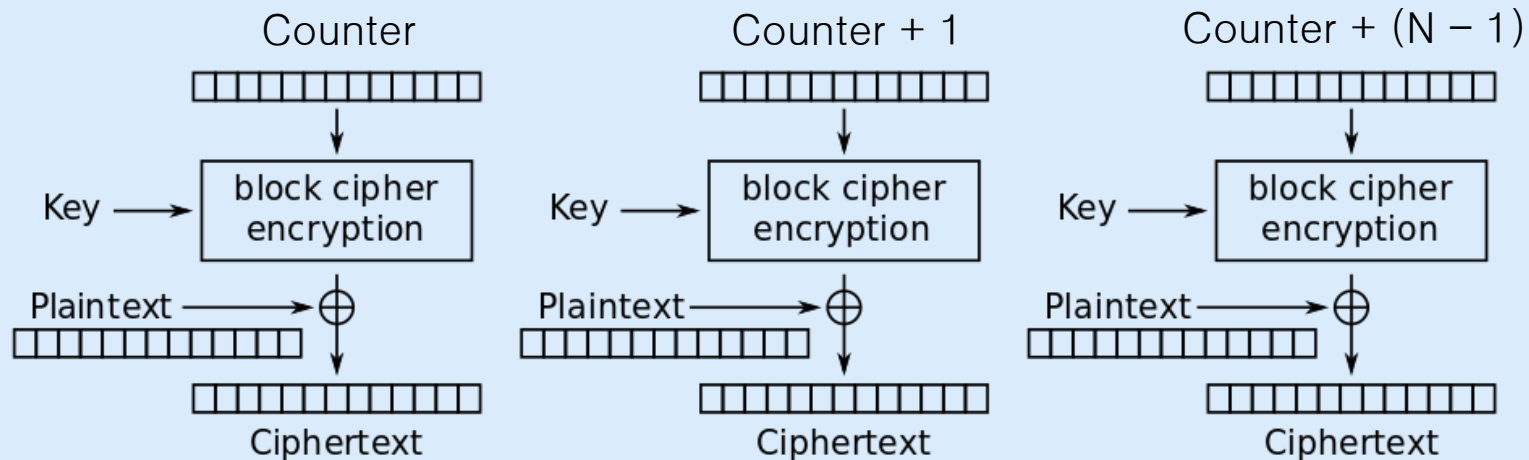


A random block to start with. Both parties can agree upon its value during key exchange.

Counter (CTR) mode



- Another one is Counter mode
- Start with a random (but agreed upon) counter value and then keep incrementing.
- This one allows encrypting blocks in parallel (unlike CBC mode)



Counter (CTR) mode encryption

Counter (CTR) mode



Questions

- How would you decrypt a message in CTR mode?
- Do you see any similarity in CTR mode and stream ciphers?

References



- Stallings CNS, chap 4, 6
- Forouzan, Cryptography & Network Security, chap 5–8
- Confusion, Diffusion and AES steps with reasoning:
<https://youtu.be/C4ATDMLz5wc>
- AES illustrated: a stick figure guide:
<http://www.moserware.com/2009/09/stick-figure-guide-to-advanced.html>
- Block cipher modes of operation – Computerphile:
<https://youtu.be/Rk0NIQfEXBA>